

# ■■ Shell Scripting Tutorial ■■

## ■ 1. What is a Shell Script?

- A shell is a command-line interpreter (bash, zsh, sh).
- A shell script is a text file with commands, usually saved with `.sh` extension.`
- Example:

```
#!/bin/bash
echo "Hello World!"
```

## ■ 2. Variables ■

- Store text, numbers, or paths in variables.

```
name="Swastik"
age=17
echo "My name is $name and I am $age years old."
```

## ■ 3. User Input ■

```
echo "Enter your favorite language:"
read lang
echo "You chose $lang"
```

## ■ 4. Conditional Statements ■■

```
if [ $num -gt 5 ]; then
echo "Number is greater than 5 ■■"
else
echo "Number is less than or equal to 5 ■■"
fi
```

## ■ 5. Loops ■

```
for i in {1..5}; do
echo "Iteration $i"
done
```

## ■ 6. Functions ■■

```
greet() {
echo "Hello, $1 ■■"
}

greet Swastik
greet World
```

## ■ 7. Command Line Arguments ■■

```
echo "Script name: $0"
echo "First argument: $1"
echo "All arguments: $@"
```

```
echo "Number of arguments: $#"
```

## ■ 8. Arrays ■

```
variable=("V" "A" "B")  
echo ${variable[0]}  
for v in "${variable[@]}"; do  
echo $v  
done
```

## ■ 9. Useful Commands ■

|          |                         |
|----------|-------------------------|
| ■ date   | Show current date/time  |
| ■ whoami | Show current user       |
| ■ ls     | List files              |
| ■ pwd    | Print working directory |
| ■ cat    | Read file contents      |

## ■ 10. Practical Example ■

**\*\*Backup Script\*\*:**

```
backup_file="/tmp/home_backup_$(date +%Y%m%d%H%M%S).tar.gz"  
  
tar -czf $backup_file $HOME  
echo "■ Backup saved to $backup_file"
```